Wed 5 Feb 2025

We've been considering the problem

solve $A\underline{x} = \underline{b}$ for $\underline{x}$

and have seen that we can think of Gaussian Elim. as factoring A into LU. (L = unit lower triangular, U = upper triangular)

We can then use the approach

1. $A\underline{x} = \underline{b}$

   $(LU)\underline{x} = \underline{b}$

   $L\underline{y} = \underline{b}$ where $\underline{y} = U\underline{x}$

2. Solve $L\underline{y} = \underline{b}$ for $\underline{y}$ using Forward Substitution

3. Solve $U\underline{x} = \underline{y}$ using Backward substitution

But! this fails for $A = \begin{bmatrix} 0 & 1 \\ 1 & 1 \end{bmatrix}$. We can rescue by interchanging rows $\longrightarrow \hat{A} = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$. similarly in $\underline{b}$

How can we remember (reasonably) all of the required row interchanges? that move 0's from the diagonal. Use permutation matrices.

Let $P_{i\,r_i}$ = the identity matrix with rows i and r_i interchanged

the ith permutation

will interchange rows i and r_i

$$= \begin{array}{c} \begin{array}{cc} \text{col } i & \text{col } r_i \end{array} \\ \begin{bmatrix} 1 & & & & & & & & \\ & \ddots & & & & & & & \\ & & 1 & & & & & & \\ & & & 0 & & & 1 & & \\ & & & & 1 & & & & \\ & & & & & \ddots & & & \\ & & & & & & 1 & & \\ & & & 1 & & & 0 & & \\ & & & & & & & 1 & \\ & & & & & & & & \ddots \\ & & & & & & & & & 1 \end{bmatrix} \end{array} \begin{array}{l} \\ \\ \\ \leftarrow \text{row } i \\ \\ \\ \\ \leftarrow \text{row } r_i \\ \\ \\ \\ \end{array}$$

= the identity matrix with columns i and r_i interchanged.

Multiplying a matrix B on the <u>left</u> by P_{i,r_i}

$$P_{i,r_i} B$$

has the effect of interchanging rows i and r_i in B $(R_i \Leftrightarrow R_{r_i})$

e.g. $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix} = \begin{bmatrix} a & b & c \\ g & h & i \\ d & e & f \end{bmatrix}$ $\quad R_3 \leftrightarrow R_2$

Multiplying a matrix B on the <u>right</u> by P_{i,r_i}

$$B P_{i,r_i}$$

has the effect of interchanging columns i and r_i in B

e.g. $\begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix} = \begin{bmatrix} a & c & b \\ d & f & e \\ g & i & h \end{bmatrix}$

$C_i \leftrightarrow C_{r_i}$

<u>Fact</u>: $P_{i,r_i}^{-1} = \ldots$ to undo $R_i \Leftrightarrow R_{r_i}$ do $R_i \Leftrightarrow R_{r_i}$,

$= P_{i,r_i}$

also $= P_{i,r_i}^T$ since P_{i,r_i} is symmetric

To add row interchanges to our Gauss Elimination algorithm, we need to permute rows with P_{i,r_i} <u>before</u> applying each Gauss Transform to remove 0's from the diagonal

Suppose we have eliminated to get

$$\begin{bmatrix} \times & \times & \times & \times & \times \\ 0 & \times & \times & \times & \times \\ 0 & 0 & \times & \times & \times \\ 0 & 0 & \times & \times & \times \\ 0 & 0 & \times & \times & \times \end{bmatrix}$$

$= 0$?
$\neq 0$?

We will want to choose the interchange row r_i to be $\geq i$ so that we do not move 0's onto the diagonal in the previously eliminated part and do not mess up structure.

Take $r_i = i$ when i,i entry in eliminated matrix is nonzero. ($P_{i,r_i} = I$)

$r_i > i$ when i,i entry in eliminated matrix is 0.

Our modified Gauss Elimination process becomes

$$M_{n-1} P_{n-1,r_{n-1}} \cdots M_2 P_{2,r_2} M_1 P_{1,r_1} A = U$$

write out this way

Let's simplify the notation and write $\boxed{M_{n-1} P_{n-1} M_{n-2} \cdots M_2 P_2 M_1 P_1 A = U}$

next: how to simplify

This doesn't look like an LU factorization any more. Let's try to isolate the Gauss Transform matrices and the Permutation matrices.

Suppose we had $M_2 P_2 M_1 P_1 A$

mult by P_2 on the left interchanges rows 2+3

with $P_2 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$ $M_1 = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 3 & 0 & 1 \end{bmatrix}$

then $P_2 M_1 = \begin{bmatrix} 1 & 0 & 0 \\ 3 & 0 & 1 \\ 2 & 1 & 0 \end{bmatrix}$

This is not the correct form for a Gauss Transform Matrix. But if we interchanged columns 2 and 3 it would be. How can we do that?

mult by P_2 on the right interchanges columns 2+3.

$$P_2 M_1 P_2 = \begin{bmatrix} 1 & 0 & 0 \\ 3 & 1 & 0 \\ 2 & 0 & 1 \end{bmatrix} = \hat{M}_1$$

This looks like a Gauss Transform matrix. Call it $\hat{M}_1$

Since $P_2^{-1} = P_2 \xRightarrow{P_2 P_2 = I}$, we can write

$$M_2 P_2 M_1 P_1 A = M_2 (P_2 M_1 P_2) P_2 P_1 A$$
$$= M_2 \hat{M}_1 P_2 P_1 A$$

We have collected the M's together and the P's together.

Proceeding in this fashion for larger n gives

$$M_{n-1}P_{n-1}\, M_{n-2}P_{n-2} \cdots P_3 M_2 P_2 M_1 P_1 A = U$$

$P_2 P_2$

$\hat{M}_2$ not a GT matrix is lower triangular

$$M_{n-1}P_{n-1}\, M_{n-2}P_{n-2} \cdots \underbrace{P_3 M_2 \hat{M}_1}_{\hat{M}_2} P_2 P_1 A = U$$

$P_3 P_3$

Continuing in this way, we arrive at the form

$$\hat{M}_{n-1}\, P_{n-1}P_{n-2} \cdots P_1 A = U.$$

$\hat{M}_{n-1}$ not a GT matrix but is lower triangular

this is a lower triangular matrix that includes all of the multipliers, possibly moved around

Let $P = P_{n-1}P_{n-2} \cdots P_2 P_1$.

$L^{-1} = \hat{M}_{n-1}$

Then we have $L^{-1} P A = U$ or $PA = LU$.

Our algorithm to solve $A\underline{x} = \underline{b}$ becomes:

1. Compute P, L, U such that $PA = LU$
2. Since $A\underline{x} = \underline{b}$, $PA\underline{x} = P\underline{b}$

   so solve $L\underline{y} = P\underline{b}$ for $\underline{y}$

   solve $U\underline{x} = \underline{y}$ for $\underline{x}$.